

### 6.3.3 Electromagnetism

| Learning outcomes                                                                                                               | Additional guidance |
|---------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>                                   |                     |
| (a) magnetic flux $\phi$ ; the unit weber; $\phi = BA \cos \theta$                                                              |                     |
| (b) magnetic flux linkage                                                                                                       |                     |
| (c) Faraday's law of electromagnetic induction and Lenz's law                                                                   |                     |
| (d) (i) e.m.f. = – rate of change of magnetic flux linkage; $\mathcal{E} = - \frac{\Delta(N\phi)}{\Delta t}$                    | M3.9                |
| (ii) techniques and procedures used to investigate magnetic flux using search coils                                             |                     |
| (e) simple a.c. generator                                                                                                       |                     |
| (f) (i) simple laminated iron-cored transformer; $\frac{n_s}{n_p} = \frac{V_s}{V_p} = \frac{I_s}{I_p}$ for an ideal transformer | M0.3                |
| (ii) techniques and procedures used to investigate transformers.                                                                |                     |